

THEORY OF COMPUTATION (CS302)

Unit-wise Syllabus Breakdown

Unit 1 - Finite Automata

Topics Covered:

- Introduction to Automata Theory
- Deterministic Finite Automata (DFA)
- Non-Deterministic Finite Automata (NFA)
- State Diagrams
- Transition Tables
- Applications of Finite Automata

Unit 2 - Regular Languages

Topics Covered:

- Regular Languages
- Regular Expressions
- Finite Automata and Regular Expressions Relationship
- Pumping Lemma for Regular Languages
- Applications of Regular Languages

Unit 3 - Context-Free Grammars

Topics Covered:

- Context-Free Grammars (CFG)
- Derivations
- Parse Trees
- Ambiguous Grammars
- Grammar Simplification
- Chomsky Normal Form (CNF)

Unit 4 - Pushdown Automata

Topics Covered:

- Pushdown Automata (PDA)
- Stack Operations
- Acceptance by Final State
- Acceptance by Empty Stack
- PDA Construction
- Equivalence of PDA and CFG

Unit 5 - Turing Machines & Decidability

Topics Covered:

- Turing Machines
- Recursive Languages
- Recursively Enumerable Languages
- Decidability
- Undecidability
- Halting Problem
- Church-Turing Thesis

Exam Preparation Tips

- Practice DFA and NFA construction regularly.
- Learn conversions between automata models.
- Focus on Pumping Lemma proofs.
- Practice CFG derivations and CNF conversion.
- Understand Turing Machine design and decidability concepts.

Summary

This document provides a structured unit-wise overview of the Theory of Computation syllabus and highlights important examination topics.